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Palaeomagnetism of the Kvamshesten and Hornelen Old Red Sandstone Basins, western Norway

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SUMMARY

Folded Old Red Sandstone in western Norway records at least four magnetization events which testify to a prolonged thermochemical history spanning mid-Palaeozoic to Mesozoic time. Sediments in the Hornelen and Kvamshesten basins carry a post-tectonic remanence which is correlated with the late Palaeozoic Kiaman superchron. Many of the overprints carried by mid and Lower-Palaeozoic rocks in Norway are reported to be significantly younger than this, and might be related to Mesozoic fault activity in the area. A second overprint identified locally in the Kvamshesten basin, adjacent to one of the present margin faults, supports the assertion.

A further remanence, the integrity of which has not been previously recognized, is probably Devonian in age. Its precise age in relation to late Caledonian folding is uncertain, and therefore various geological models based on the data are offered. The simplest of these is that the remanence is post-tectonic and that Britain and Norway attained their present relative configuration in Devonian times. Nevertheless, much of the recently obtained Middle Palaeozoic palaeomagnetic data for Norway perpetuate the longstanding discrepancy between Norwegian and British apparent polar wander (APW). Uncertainty in time calibration of the paths cannot be the sole explanation for this. The difference might be founded in post mid-Devonian tectonism (in Norway?) or, perhaps more likely, in our understanding of the *detailed* APW of the relevant continents.

Key words: basin analysis, Norway, palaeogeography, palaeomagnetism, tectonomagnetism.

INTRODUCTION

There is a growing similarity between the Palaeozoic continental reconstructions of the North Atlantic bordering continents (Livermore, Smith & Briden 1985; Scotese, van der Voo & Barrett 1985; Briden *et al.* 1989). Now that the apparent polar wander paths for Scotland and North America are showing a significant similarity in form, the position of Baltica in Lower to Mid-Palaeozoic time is perhaps one of the most fundamental and outstanding of the remaining problems in reconstruction of the North Atlantic region. This is because there are currently few reliable data of Cambrian to Devonian age from either Fennoscandia or the Russian Platform (Smethurst & Khramov 1989). Recent reviews of palaeomagnetic data from those areas have been offered by Pesonen *et al.* (1989), Smethurst & Khramov (in preparation) and Khramov, Petrova & Pechersky (1981). Notwithstanding intensive study in recent years, Norwegian apparent polar wandering remains controversial. This study, based on Devonian Old Red Sandstone facies sediments in western Norway (Fig. 1), was undertaken with a view to contributing new data to the currently sparse database. A brief review of 'Devonian' palaeomagnetic data from Norway is made, and the implications of the new data presented here will be discussed in the context of continental scale palaeogeographies.

One further objective of this study was to provide an additional constraint on the tectonic evolution of the Hornelen and Kvamshesten sedimentary basins in western Norway (Fig. 1). A number of contrasting models have been proposed for the development of the Devonian basins (including the Håsteinen, and Solund massifs). The margins

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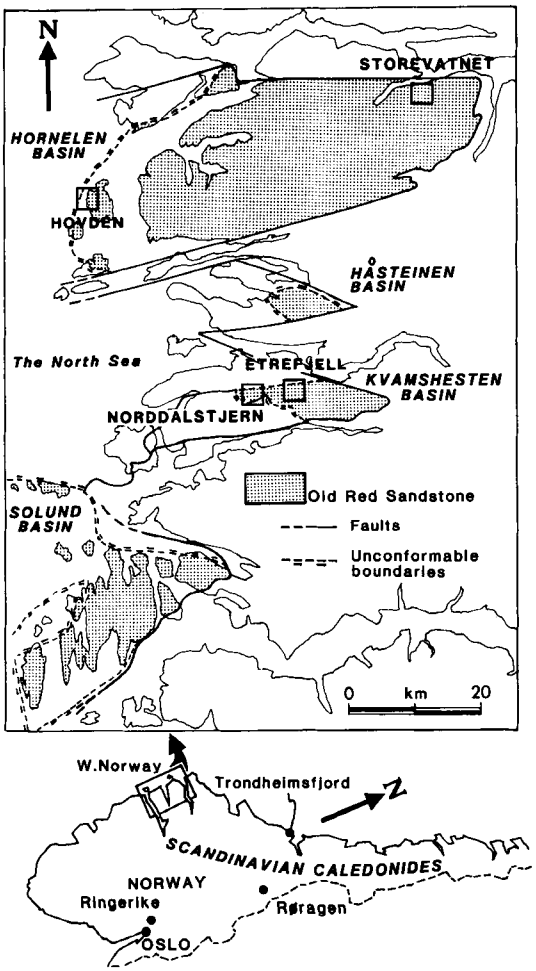


Figure 1. The Old Red Sandstone basins of Norway (western Norway, Trondheimsfjord, Røragen). Sampling regions Etrefjell Norddalstjern, Hovden and Storevatnet are indicated.

of the four Devonian outliers are a complex arrangement of unconformities and high- and low-angle detachment(s). They enclose folded Devonian sediments; some are inverted. Palaeomagnetic analysis concentrated near the basin margins has already proven itself as a powerful tool for constraining the chronology of deformation/rotation of the basinal and adjacent rocks (Torsvik *et al.* 1986, 1989c), yet it should be noted that thermal and chemical magnetic overprinting is likely in the rocks near to the tectonic basin margins and that further magnetotectonic information might be obtained from rocks in the interior parts of the basins. This study is a palaeomagnetic analysis of megascopically uncleaved Devonian sediments, where practical, away from the tectonic basin margins, where early/primary magnetic remanence could survive. Consequently this analysis provides additional and important, but complex, new information on the early, perhaps syn-depositional, tilting/rotation history of the sediments.

GEOLOGY

Four main Devonian sedimentary basins lie in the Sogn and Fjordane region of western Norway (Fig. 1). The basins contain thick sequences of non-marine conglomerates, sandstones and siltstones in varying proportions which unconformably overlie Lower Palaeozoic metamorphic 'basement' rocks. Their age is demonstrated by Middle Devonian plant and fish remains in the youngest sandstones, and similar early Devonian fossils in the Solund basin (Kolderup 1915; Kiaer 1918; Hoeg 1936; Jarvik 1949; Steel & Aasheim 1978). Sediments within the Hornelen and Kvamshesten basins comprise a stream-flow dominated alluvial fan and fan-delta bodies, recording a sedimentary cyclicity on several scales. The Hornelen basin fill consists of small-scale fan deposits along the northern and southern faulted margins and a larger scale east to west axial fan-delta system. Fine-grained flood basin and lacustrine deposits associated with the axial system are situated close to the northern marginal fan bodies. Alluvium in the Hornelen basin shows a large-scale upward coarsening cyclicity (Steel *et al.* 1977) with smaller scale, superimposed, upward fining units. This is taken to be a response to intermittent basin subsidence and easterly depocentre migration (Steel 1976).

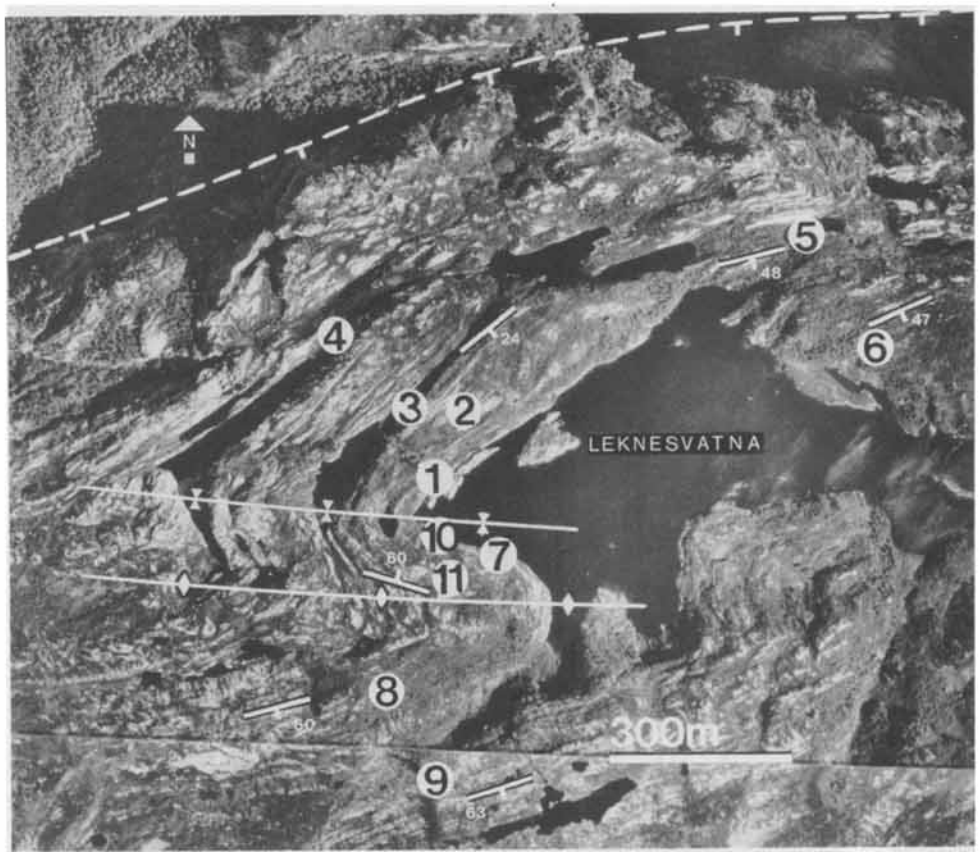
The Kvamshesten basin is fringed with fanglomerates which dominate in the southern half of the basin and thin to the north, giving way to an E–W tract of braided terminal floodbasin and lacustrine deposits, which show an alternation between red siltstones and green sandstones (the Heilefjell Fm.; Steel, Siedlecka & Roberts 1985). The sedimentology of the Hornelen basin is described in detail by Steel & Aasheim (1978), Steel *et al.* (1977) and of the Kvamshesten basin by Høisethær (1971), Bryhni & Skjerlie (1975), and general summaries for the Devonian of western Norway have been offered by Bryhni (1964), Steel (1976), Steel & Gloppen (1980) and Steel *et al.* (1985).

The Solund and Kvamshesten basins are clearly underlain by a major low-angle detachment, outcropping on their eastern margins. The detachment was first interpreted as a thrust. Subsequently Hossack (1984) suggested that the four Devonian basins formed above, and now overlie, a single listric normal fault, with steep lateral ramps, accommodating late orogenic extensional collapse. The argument for normal movement on the basal detachment(s) has been extended by Norton (1986), Chauvet & Brunel (1988), Chauvet & Séranne (1988) and Séguret *et al.* (1989). It seems likely that the detachment, whether single or composite, has its origins as a Scandian thrust structure (Hossack 1984) and was later reactivated as a normal fault during orogenic collapse, accommodating the development of the syn-tectonic sedimentary basins. The attribution of observed deformation in and around the massifs to (1) syn-sedimentary extension and (2) a late Devonian to early Carboniferous post-depositional tectonic episode (Svalbardian/Solundian orogeny; Sturt 1983; Roberts 1983; Torsvik *et al.* 1989c) is still an issue for debate.

The crystallinity of illite and phengite (see Kisch 1987; Padan, Kisch & Shagam 1982) from the Devonian clastics establishes an upper anchizone metamorphic grade (*c.* 200°–300 °C; Torsvik *et al.* 1986).

DISTRIBUTION OF PALAEOMAGNETIC SAMPLING SITES

Palaeomagnetic core samples were collected from both the Kvamshesten and Hornelen basins. The majority of samples from the Kvamshesten basin are from 11 sites in folded red sand/siltstones of the Heilefjell Fm. (Fig. 1, 2a). This

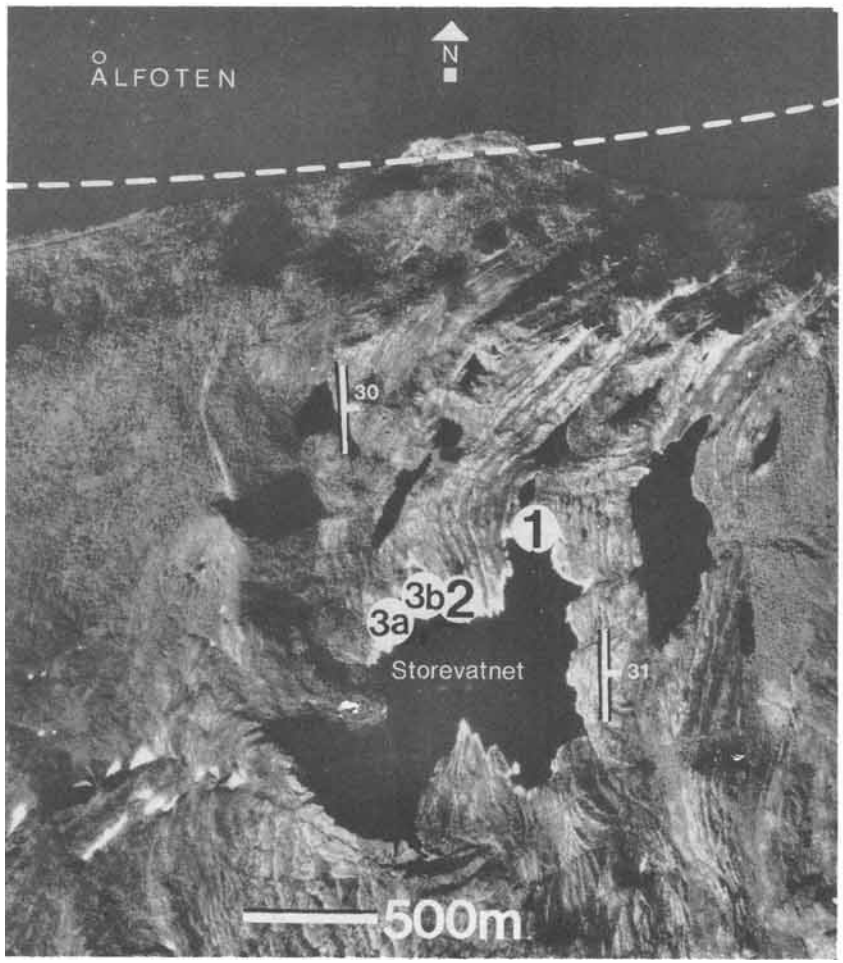


(a)

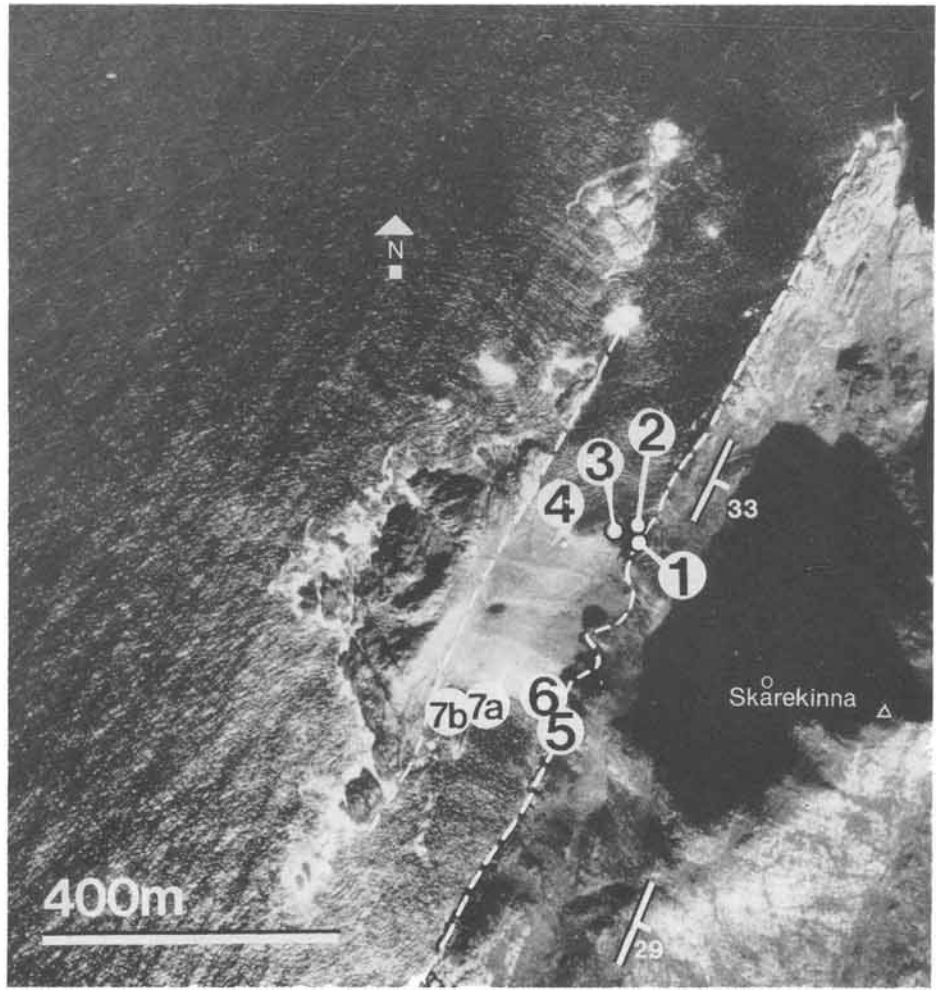


(b)

Figure 2. Aerial photographs showing sampling sites and some geological features. Kvamshesten Heilefjell Fm.: (a) sites E1–11 at Etrefjell, (b) sites N1–4 at Norddalstjern. Hornelen: (c) Kvannhovdvika Fm. sites H1–7, Hovden, (d) sites S1–3 at Storevatnet. Aerial photographs courtesy of Fejellanger Widerøe a.s.



(c)



(d)

Figure 2. (*continued*)

sampling region is here termed Etrefjell (and the magnetic properties of these rocks are referenced by the subscript E). Redbeds of the Heilefjell Fm. were sampled at a further four sites at the western end of the basin (Norddalstjern: N; Fig. 1, 2b; Bryhni & Skjerlie 1975).

The Hornelen basin was also sampled in two areas. The first area is Hovden island (H) where red and green sand/siltstone samples were collected from seven sites in the Kvannhovdvika Fm., which outcrops on the western shore of the island (Fig. 1, 2c; Larsen *et al.* 1981). The second sampling area, called Storevatnet (S; Fig. 1, 2d; Pollard, Steel & Undersrud 1982; Larsen & Steel 1978), lies in the northeast of the Hornelen massif. In this area oriented block samples were collected at four stratigraphic levels in a sequence of red and green sand/siltstones.

The magnetic remanence labelling scheme for all sampling areas is summarized below.

Stability		Low (L) <200 °C	Intermediate (I) *c.* 200–630 °C	High (H) >600 °C
Hovden	$(_H)$	L_H	I_H	H_H
Storevatnet	$(_S)$	L_S	I_S	H_S
Etrefjell	$(_E)$	L_E	I_E	H_E
Norddalstjern	$(_N)$	L_N	I_N	H_N

e.g. intermediate T_b at Hovden and Storevatnet is denoted $I_{H,S}$, and low and intermediate at Hovden is denoted L, I_H.

PALAEOMAGNETIC RESULTS FROM THE KVAMSHESTEN BASIN

The Heilefjell Fm. at Etrefjell: E

Natural Remanent Magnetization (NRM) intenstities of the majority of redbed samples lie between 2.0 and 10.0 mA m^{-1}, and of the green sandstones, between 0.1 and 5.0 mA m^{-1}. NRM components were identified using both progressive Thermal and Alternating Field (AF) demagnetization techniques. The thermal technique effectively demagnetized samples of all lithologies, although mineralogical alteration was detected following high-temperature treatments (>630 °C). The new (high magnetic susceptibility) phase, with Curie point at *c.* 570 °C, is probably magnetite. Linear/planar elements in demagnetization trajectories were isolated using the LINEFIND algorithm of Kent, Briden & Mardia (1983).

Total NRM directions for all 69 demagnetized samples are given in Fig. 3(a). The majority of NRM directions are dispersed between a southerly to southwesterly downwards and a south and upwards trend. The remaining NRM directions are widely scattered in the eastern hemisphere of the projection.

Most demagnetized samples were found to carry three components of NRM (Fig. 4). They are, in order of increasing blocking temperature (T_b)/coercive force stability range, labelled L_E (low: <200 °C), I_E (intermediate: 150–650 °C) and H_E (high: 610–680 °C). Remanence components are assigned a label for descriptive purposes only. It is *not* suggested that all components with the same label necessarily have the same age and/or mode of origin.

Although the L_E component is poorly defined, it is directed steeply downwards and is regarded as recent viscous remanent magnetization (VRM). Almost all component I_E remanences are directed to the south/southwest with shallow to intermediate negative inclinations. Site level I_E component directions are given in Fig. 5(a) and are listed in Table 1. H_E component directions were reliably determined for a number of sites using the linear structure in demagnetization trajectories alone (LINEFIND). At other sites it was necessary to combine planar, linear and stable endpoint data to derive the direction of H_E. At some of these other sites the ratio of H_E remanence directions identified, to samples demagnetized, was low. A four-level ranking system has been applied to site level H_E component directions which reflects the nature of the sample level data from which they are each derived (Table 2 and Fig. 6). Those site level directions which are based only on clearly defined linear elements in demagnetization trajectories are ranked 1. Those data based on few intersecting great circle data and/or weak linear trends in demagnetization data rank 4.

The most frequently identified, and acceptable, H_E component trend is to the southwest with an intermediate to steep downward inclination (Fig. 6). This directional trend is exclusive to samples from sites E1, 2, 5, 8 and 9. These data

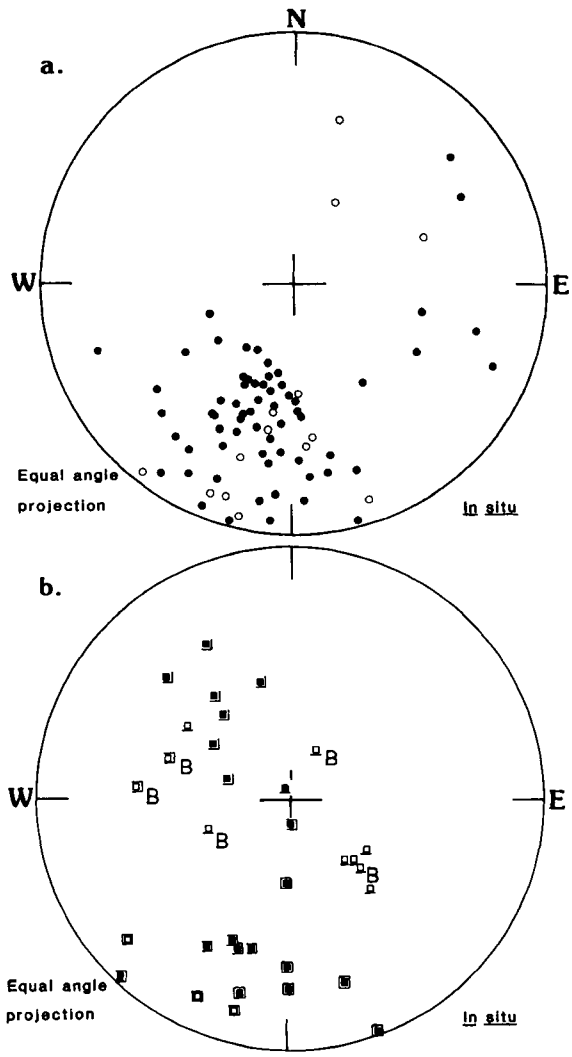


Figure 3. NRM directions of samples from (a) Etrefjell and (b) Norddalstjern area, Kvamshesten basin. In (b) 'B' denotes those samples with high NRM intensities. The number of strokes adjacent to each symbol (1–4) correspond to site number (N1–4).

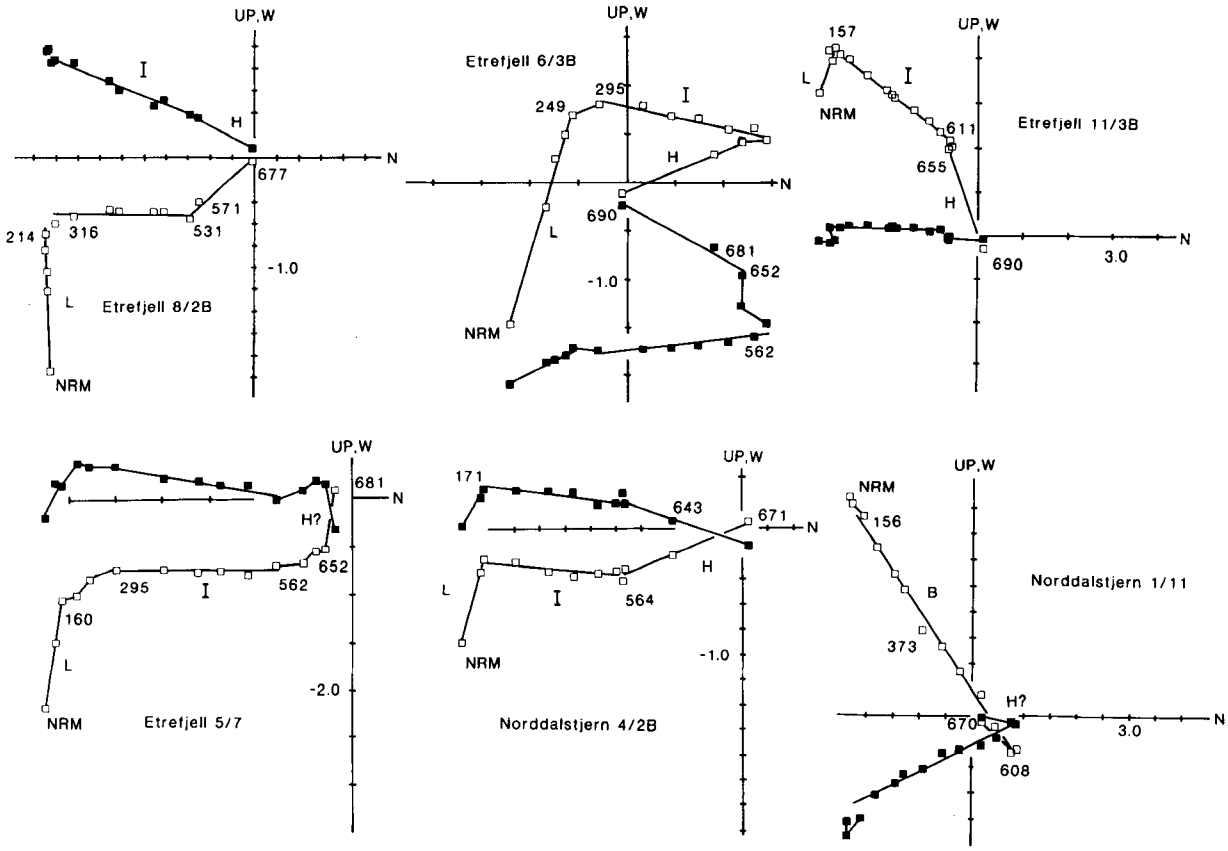


Figure 4. Orthogonal plots of demagnetization data: the Etrefjell and Norddalstjern areas, Kvamshesten basin. Open (resp. closed) symbols are projections in the vertical (resp. horizontal) plane. Intensities are in mA m^{-1} and treatments in °C.

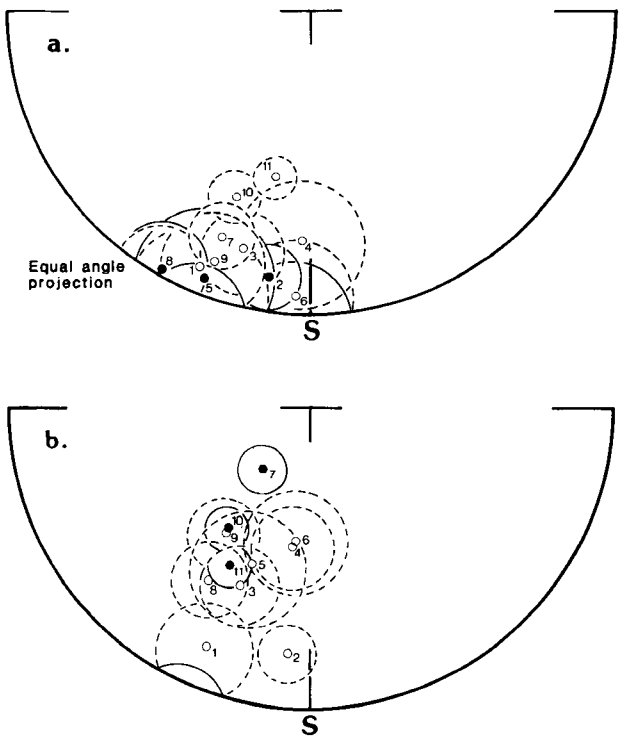


Figure 5. Site level I component directions (a) *in situ* and (b) 'unfolded' from the Etrefjell area. Equal angle projection; open (resp. closed) symbols are in the upper (resp. lower) hemisphere. Site numbers and limits of 95 per cent confidence are shown.

have a moderate to low within site scatter (Table 2), rank highly and are unambiguous. A number of other directional trends were implied by the data. Those trends, shown in Fig. 6 and listed in Table 2, are based on few reliable sample level data and are not acceptable for further palaeomagnetic analysis. A tectonic explanation for the dispersal of all high-ranking data is considered later.

THE HEILEFJELL FORMATION AT NORDDALSTJERN: N

The magnetic properties of the majority of samples from Norddalstjern are similar to those for the nearby and laterally equivalent strata at Etrefjell. NRM directions of all 33 demagnetized samples are shown in Fig. 3(b). The NRM of samples from sites N3 and 4 is directed to the south/southwest with shallow to intermediate downward inclinations. NRM directions from sites N1 and 2 are southeast and upward, and northwest and downward respectively (antipodal). NRM intensities lay between 1.0 and 5.0 mA m^{-1}. Some samples from sites N1 and 3 with intermediate to steeply upward NRM directions had intensities of 100–400 mA m^{-1}.

NRM components L_N, I_N and H_N (low, intermediate and high stability) were isolated at sites N3 and 4 (Fig. 4). A fourth characteristic NRM component was identified in samples from sites N1 and 2 (and some samples from site N3; Fig. 4) which has a broad stability range, between room temperature and 650 °C, and has therefore been given the

Table 1. Site level *I* component directions for the Heilefjell Fm., Etrefjell area.

Site	In situ		1/3 'unfolded' Ł		2/3 'unfolded' Ł		Completely 'unfolded'†		*N*	*k*	α_{95}
	Dec.	Inc.	Dec.	Inc.	Dec.	Inc.	Dec.	Inc.			
E1	202.6	−4.6	198.0	−13.9	198.8	−12.2	202.8	−8.5	6	41.5	10.5
E2	188.5	6.7	185.7	0.9	182.6	−4.8	184.7	−11.2	6	79.7	7.6
E3	195.3	−11.3	192.2	−17.5	189.2	−23.7	200.3	−25.7	6	40.8	10.6
E4	181.6	−15.2	174.3	−25.6	166.2	−34.2	185.8	−40.3	5	21.9	16.7
E5	201.2	2.6	194.8	−9.6	188.8	−22.1	200.0	−31.1	7	14.8	16.2
E6	182.6	−3.8	177.2	−14.7	171.6	−25.6	186.9	−40.2	5	33.8	13.4
E7	200.6	−12.0	194.7	−17.6	189.9	−20.3	216.1	60.9	6	55.6	9.1
E8	109.5	1.2	203.3	−9.4	198.7	−20.8	209.8	−23.7	7	36.5	10.1
E9	200.3	−6.3	195.0	−18.6	191.6	−31.5	212.7	−36.3	7	27.9	11.6
E10	201.1	−22.5	208.2	−7.0	216.5	8.2	213.8	38.1	6	94.9	6.9
E11	191.3	−30.3	197.4	−13.8	205.6	1.9	206.3	28.1	5	142.8	6.4

Dec., Inc. = remanence declination and inclination; N = number of samples; k = Fisher's (1953) precision parameter; α_{95} = half angle of the cone of 95 per cent confidence about the mean direction; Ł = 'unbuckled' only; † = 'unplunged' and 'unbuckled'. *In situ* and 'unfolded' directions are given in Fig. 5. Structural data are given in Table 2.

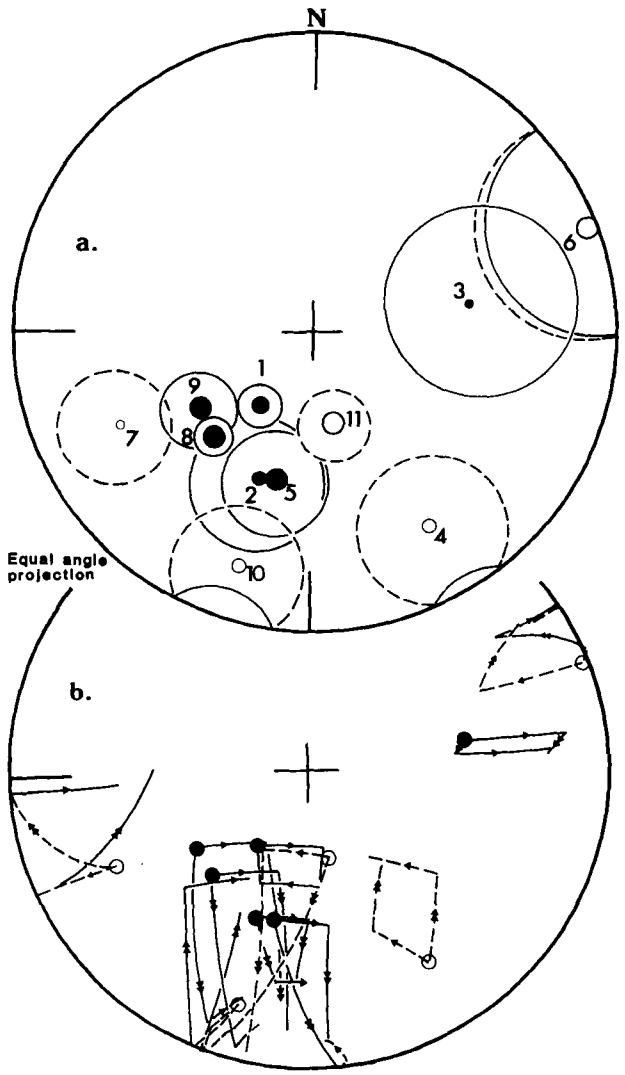


Figure 6. (a) Site level *H* component directions for the Etrefjell area. Large (resp. small) symbols denote well-resolved (resp. 'noisy') sample level data (see text). (b) Lines with a single (resp. double) arrow denote the effect of 'unplunging' (resp. 'unbuckling'). Site numbers are indicated.

label B_N (broad T_b spectrum). The NRM of samples carrying remanence B_N was usually single component, and often of high intensity. A number of samples from site N1, however, where B_N is well developed, carry an additional low-intensity component stable between 650° and 680 °C. This component is labelled H_{N1} but has a narrower stability range than H elsewhere.

L_N and I_N component directions (Fig. 4) coincide with those from Etrefjell samples (L_E, I_E). H_N component directions are to the south–southwest with shallow downward inclinations. This trend is shallower than the well-defined H_E trend for Etrefjell sites E1, 2, 5, 8 and 9. H_{N1} is directed downward to the north. B_N component directions are generally to the southeast and up (at site N1) and northwest and down (site N2). No counterpart for B_N was identified in the Etrefjell region. Comparison of total NRM and site level NRM component directions for Norddalstjern (Fig. 3b and Fig. 7; Table 3) illustrates the prominence of I_N and H_N at sites N3 and 4 and B_N at sites 1 and 2.

THE FOLD TEST

Palaeomagnetic sampling in the Kvamshesten basin was carried out around plunging folds. For the location of sites in relation to fold axes refer to Fig. 2(a, b). Bedding orientations measured at the sampling sites are listed in Tables 2 and 3. They define fold axes plunging at approximately 27°–094° in the Etrefjell area and 26°–072° at Norddalstjern.

A necessarily simplistic approach has been adopted to analyse the possible effect of fold development on potential pre- or syn-folding remanence component directions ($H_{E,N}$ and $I_{E,N}$). The effect of possible folding histories on remanence directions has been modelled by 'opening' (or 'unbuckling') and 'unplunging' the folds/remanence directions by various amounts. In doing so it is practical to assume that (1) fold hinges have not migrated, (2) fold hinge angles have never been smaller than at present (or >180°) and (3) that the original angular relationship between remanence and local bedding has not been altered by internal strain. This third assumption is validated by

Table 2. Site level *H* component directions for the Heilefjell Fm., Etrefjell area (sites are not listed in stratigraphic order).

Site	Bedding		Fold axes		*In situ*		'Unfolded'†		*N*	*k*	α_{95}	Reliability classification*
	Strike	Dip	Azimuth	Plunge	Dec.	Inc.	Dec.	Inc.				
E1	032	29	094	27	215.0	56.3	174.5	48.0	5	103.9	7.5	2
E2	051	27	094	27	199.8	34.6	184.4	18.6	7	9.9	20.2	3
E3	046	33	094	27	078.8	34.6	085.5	13.8	4	11.0	29.0	4
E4	059	31	094	27	148.9	−16.1	142.8	−47.2	4	24.9	18.8	3
E5	069	48	094	27	193.3	35.3	176.2	−6.1	6	16.2	17.2	1 Ł
E6	058	47	094	27	068.0	−2.2	054.2	−9.0	4	13.2	26.3	1
E7	293	73	094	27	243.9	−19.3	270.7	35.9	4	34.9	15.8	4
E8	058	58	094	27	222.5	39.0	185.4	8.1	6	117.2	6.2	1 Ł
E9	058	63	094	27	235.5	41.4	186.6	15.6	7	20.0	13.9	1 Ł
E10	298	61	094	27	196.8	−11.8	205.5	47.8	6	19.4	15.6	3
E11	280	59	094	27	167.0	−57.8	193.3	2.3	6	26.8	13.2	1 Ł

Notation as defined in Table 1 except † = 'unfolding' incorporating 'unplunging' followed by 'unbuckling'; * = see text for explanation of ranking; Ł = directions selected for the fold test in Table 4(a). See Fig. 2(a) for position of sites on the folds; *in situ* and 'unfolded' directions are plotted in Fig. 6.

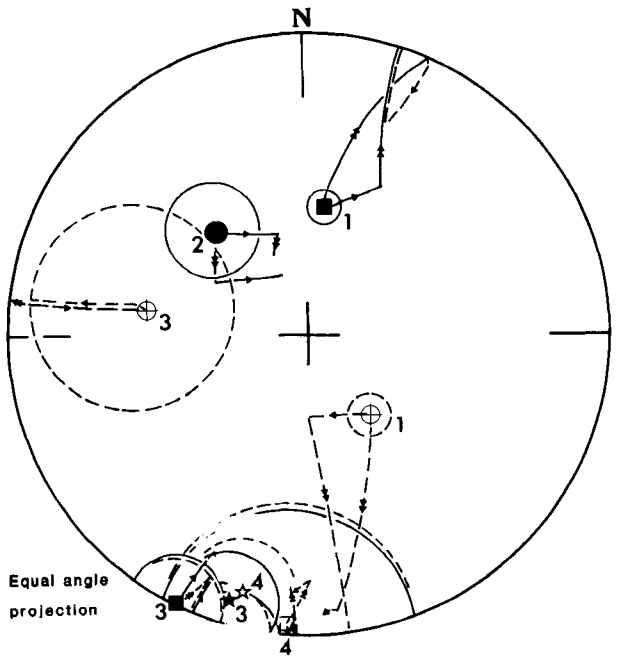


Figure 7. Site level *I* (stars), *H* (squares) and *B* (circles) component directions for the Norddalstjern area. Site numbers are indicated. Conventions are as in Fig. 6.

magnetic fabric (anisotropy of magnetic susceptibility) studies. Within the sampling area the sediments have a weak bedding-parallel magnetic foliation; a common feature in sedimentary rocks, which can be wholly attributed to sedimentary and/or diagenetic processes.

Figure 7 is an equal angle projection of *in situ* H_E component directions. The nearly closed domains therein, bounded by four small circle elements, enclose the possible former syn- to pre-folding orientations for the *in situ* data which lie at one of their 'corners'. The small circle margins to the domains indicate the directional effect of complete 'unplunging' followed by 'unbuckling' and also the converse. Comparison of such domains for coeval remanences from various parts of a plunging fold might give some qualitative indication of fold morphology at the time of remanence acquisition. It should be borne in mind that similar domains for a steeply plunging tight fold might be 'broad' and 'long' and that the error bounds on remanence directions must also be taken into account. Hence there might be a wide range of possible 'unfolding paths' for each data element. Consequently, in these circumstances, an incremental/continuous fold test should be applied with caution. For example, if the analyst believes it is possible to freely infer remanence acquisition at any former fold shape within the confines of the aforementioned assumptions, some improvement in remanence grouping is almost inevitable but not necessarily meaningful.

In the Kvamshesten basin the geometry of the folds and remanence dictate that 'unplunging' rotations generally have a larger effect on $H_{E,N}$ directions than on $I_{E,N}$ [Figs 6 and 5(a, b)]. On the other hand 'unbuckling' tends to have a large rotational effect on both $H_{E,N}$ and $I_{E,N}$. Since, for $I_{E,N}$, the rotational effect of 'unbuckling' is considerably larger than that of 'unplunging', an incremental 'unbuckling' approach was adopted for the $I_{E,N}$ component fold test. Conversely, $H_{E,N}$ component directions have indeterminate 'unfolding' paths and therefore a simpler conventional *in situ* against 'unfolded' test was used for $H_{E,N}$ directions.

COMPONENT I

In situ and 'unfolded' site level $I_{E,N}$ directions are given in Figs 5(a, b) and 7 and Tables 1 and 3. Statistically significant site level I directions were obtained for all sites at Etrefjell and two at Norddalstjern, which are clearly better grouped *in situ* than 'unfolded'. Mean I_E remanence directions were progressively rotated about plunging fold axes for the fold test (Table 1). Site level directions become increasingly dispersed upon incremental 'unfolding' (Table 4b), however the partially 'unfolded' distribution only becomes significantly worse than the *in situ* group at 2/3 or more 'unfolding' (at the 95 per cent confidence level). An immediate and slight rise in Fisher's precision parameter occurs upon 'unfolding' but it is not statistically significant and hence no inference is made regarding the feature. The fold test simply suggests that I_E is either post- or late folding in age. Qualitative comparison of *in situ* and 'unfolded' I_E and I_N directions [Figs 5(a, b) and 7] carried by rocks with contrasting bedding dip confirms this conclusion.

Table 3. Site level NRM component directions for the Heilefjell Fm., Norddalstjern area (sites are not in stratigraphic order).

Component *H*

Site	Bedding		Fold axis		*In situ*		'Unfolded'†		*N*	*k*	α_{95}
	Strike	Dip	Azimuth	Plunge	Dec.	Inc.	Dec.	Inc.			
N1	269	59	072	26	008.7	43.7	020.9	−14.7	3	534.5	5.3
N2	012	24	072	26	—	—	—	—	—	—	—
N3	005	30	072	26	207.3	0.3	203.1	11.1	6	44.5	10.2
N4	019	34	072	26	185.3	−2.4	185.6	−9.6	7	5.0	30.1

Component *I*

Site	*In situ*		'Unfolded'†		*N*	*k*	α_{95}
	Dec.	Inc.	Dec.	Inc.			
N1	—	—	—	—	—	—	—
N2	—	—	—	—	—	—	—
N3	198.6	−1.7	197.3	5.3	6	33.6	11.7
N4	194.5	−5.3	194.9	−6.9	9	16.4	13.1

Component

Site	*In situ*		'Unfolded'†		*N*	*k*	α_{95}
	Dec.	Inc.	Dec.	Inc.			
N1	143.0	−52.4	172.5	0.8	8	56.5	7.4
N2	319.2	41.5	338.9	58.0	7	16.3	15.4
N3	279.7	−33.1	277.5	−3.7	2	68.1	30.7
N4	—	—	—	—	—	—	—

Notation as in Table 1. The positions of sites around the fold are given in Fig. 2b. *In situ* and 'unfolded' directions are given in Fig. 7.

Table 4. Group of sites level statistics and fold test, Etrefjell area. (a) *H* component (using directions from E5, 8, 9, 11 in Table 2).

Overall statistics							VGP				
Mean direction:	Declination	Inclination	*N*	*R*	*k*	α_{95}	Long.	Lat.	*dm*	*dp*	*Lp*
in situ	207.6	21.4	4	2.8	2.5	74.4	Not statistically significant				
'unfolded'†	185.4	3.8	4	3.9	44.8	13.9	359.4E	−26.5	13.9	7.0	1.9S Ł

McElhinney Fold Test:		Observed *F*-ratio k2/k1	Critical *F*-ratio Level of significance	
			0.05	0.01
k2('unfolded'):	44.8	17.9		
k1(*in situ*):	2.5		4.28	8.47

(b) *I* component (using directions from E1–11 in Table 1)

							VGP				
Mean direction:	Declination	Inclination	*N*	*R*	*k*	α_{95}	Long.	Lat.	*dm*	*dp*	*Lp*
in situ	196.0	−8.8	11	10.7	33.2	8.0	346.6E	−31.7	8.1	4.1	4.4N Ł
1/3 'unfolded'*	193.0	−13.5	11	10.8	45.1	6.9	349.7E	−34.6	7.1	3.6	6.9N
2/3 'unfolded'*	191.1	−17.4	11	10.4	18.1	11.0	351.6E	−36.8	11.4	5.9	8.9N
'unfolded'†	201.2	−10.2	11	9.1	5.3	22.0	1340.4E	−31.5	22.3	11.3	5.1N

McElhinny Fold Test:		Observed *F*-ratio k1/k2–4	Critical *F*-ratio Level of significance	
			0.05	0.01
k1(*in situ*):	33.2			
k2(1/3 'unfolded'):	45.1	0.7	2.12	2.94
k3(2/3 'unfolded'):	18.1	1.8	2.12	2.94
k4('unfolded'):	5.3	6.3	2.12	2.94

Hence *in situ* directions are significantly better grouped than completely 'unfolded' directions at the 99 per cent confidence level.

Notation as in Table 1 except VGP = virtual geomagnetic pole; Long., Lat. = longitude and latitude; *dm*, *dp* = half angle of the oval of 95 per cent confidence on the VGP in the direction of and perpendicular to the magnetic meridian; *Lp* = palaeolatitude; * = partial 'unfolding' consisting of 'unbuckling' only; Ł = VGP included in Fig. 12.

COMPONENT $H_{E,N}$

The well-defined $H_{E,N}$ remanence components at E1, 2, 5, 8, 9 and 11 and N3 and 4 (Figs 6 and 7) can be reconciled by 'unfolding' them. The poorly defined (except H_{E6}) and dispersed H_E trends identified at some of the remaining sites cannot be readily brought into alignment with eachother and results from the above named sites by 'unfolding', although their magnetic properties are otherwise similar. Therefore the chrono/tectonomagnetic significance of these few data is currently unresolved. The best defined mean directions $H_{E5,8,9,11}$ are significantly better grouped 'unfolded' than *in situ* at the 99 per cent confidence level. [With a sample of Four, a substantial increase in Fisher's (1953) precision is required for significance: Table 4(a).] Furthermore, the strong directional trend defined at sites E1, 2, 5, 8 and 9 converges with directions from N3 and 4 (nearby) upon complete 'unfolding'. Hence at face value the fold test suggests a pre-/early folding age for $H_{E,N}$. It should be recognized, however, that some ambiguity is associated with this test. Firstly, a number of outlying data (e.g. at E6) cannot be explained in terms of the fold model. Secondly, if the single site level result H_{E11}, alone from one of the fold limbs (Fig. 2a), was not correlative with H_E then the fold test would be inconclusive and the overall remanence direction would be steeper and closer to a southwestward

trend, concordant with some Devonian results from Britain (Cheviot granite and lavas, Thorning 1974; Scottish Old Red Sandstone lavas, Stubbs 1958; Eday lavas, Robinson 1985; Kinghorn Lavas, Everitt & Belshe 1960) and Spitzbergen (Wood Bay Fm., Løvlie *et al.* 1984).

COMPONENT B_N

In situ B_N component directions at N1 and 2 are approximately antipodal and could record a reversal in the Earth's field. Application of 'unfolding' rotations to these data breaks down that good alignment. This strongly suggests that B_N post-dates folding.

PALAEOMAGNETIC RESULTS FROM THE HORNELEN BASIN

The Kvannhovdvika Fm. at Hovden: H

NRM intensities of the 46 demagnetized samples from the Kvannhovdvika Fm. lie between 2.0 and 15.0 mA m^{-1}. Green siltstone samples were the most weakly magnetized. NRM directions for all samples are given in Fig. 8. Both the red and green siltstones carry NRM with south to west upward and northeast up and downward directions. Between one and three NRM components were identified in 31 of the demagnetized samples (Fig. 9). NRM vector components are characteristically removed during thermal demagnetization in the ranges room temperature to 150 °C; 100° to 500–630 °C; 550° to 680 °C. These stability ranges compare closely with those of NRM components identified in the Kvamshesten basin (low, intermediate and high: $L_{E,N}$, $I_{E,N}$, and $H_{E,N}$ respectively). Consequently they are assigned the labels L_H, I_H, and H_H.

Component L_H is carried by most samples and always constitutes a small fraction of total NRM. L_H is generally directed steeply downwards yet its direction is usually poorly defined by data. This component is regarded as recent VRM.

The commonly isolated I_H component is directed to the south–southwest to southwest with shallow upward inclinations. Site level (and some sample level) I_H component directions are given in Fig. 10(a) and listed in Table 5.

Of the three NRM components, H_H is the larger. Component H_H was clearly identified in comparatively few samples, due to mineralogical alteration and associated sample magnetic instability following high thermal demagnetization treatments (>630 °C; Fig. 9). H_H component directions are consistent within sampling sites but vary between sites. Site level (and some sample level) H_H component directions are given in Fig. 10(b). The only H_H component directional trend to be identified at more than one site is northeast with shallow, generally downward inclinations.

The bimodal distribution in NRM directions of samples from the Kvannhovdvika Fm. clearly reflects the occurrence of varying proportions of the underlying I_H and H_H NRM components.

The Siltstones at Storevatnet: S

A total of 17 samples of red siltstone and 15 of green siltstone were demagnetized. Thermal demagnetization proved the most effective method for resolving NRM components, although some mineralogical alteration was detected through a rise in bulk susceptibility following high-temperature treatments. Optimum resolution of remanence components carried by the redbeds was obtained by first treating samples with alternating fields up to 50 mT (peak) and then thermally demagnetizing them.

Total NRM intensities were between 0.4 and 5.0 mA m^{-1}

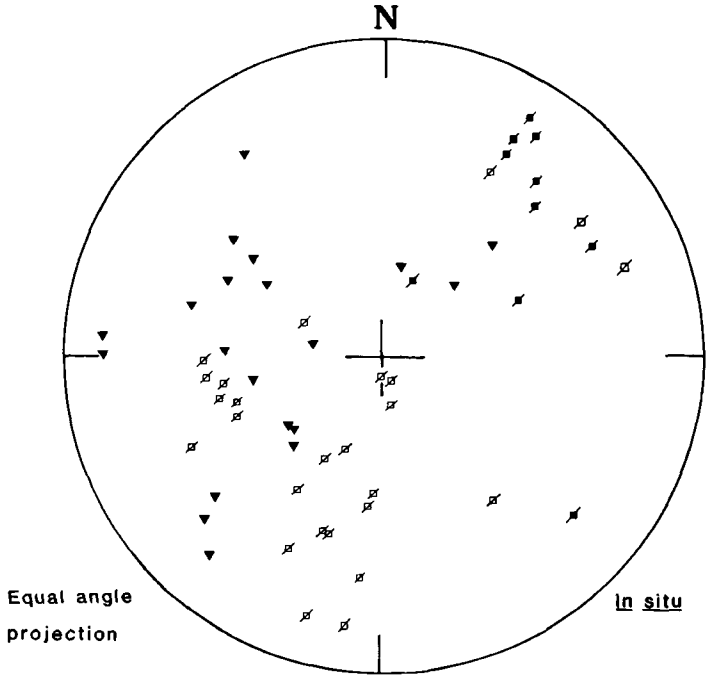


Figure 8. NRM directions of samples from the Hornelen basin. Squares: Hovden, triangles: Storevatnet.

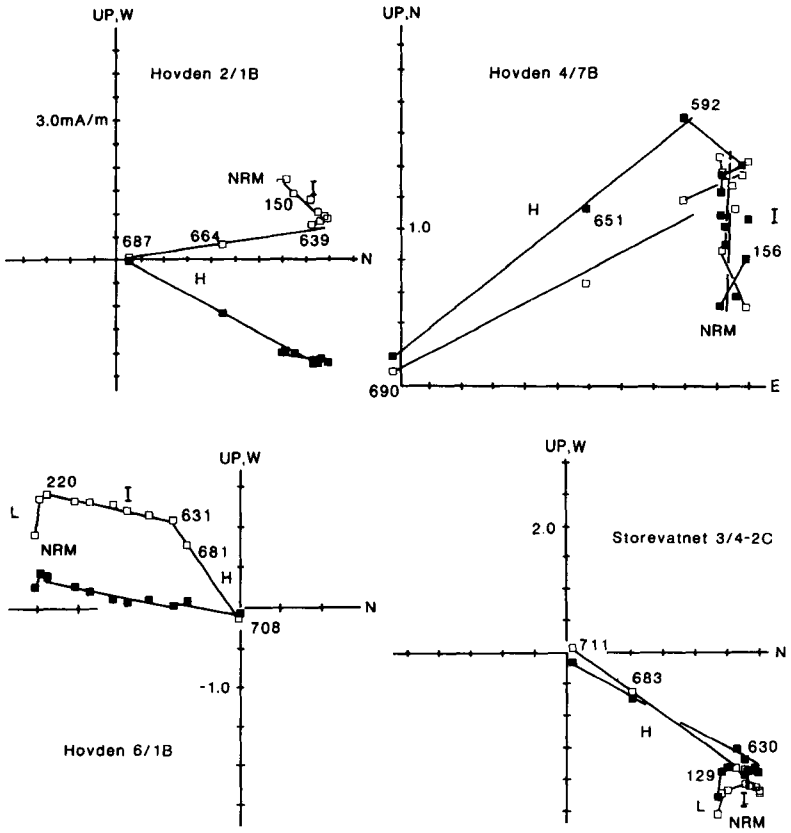


Figure 9. Orthogonal plots of demagnetization data for samples from Hovden and Storevatnet. Conventions are as in Fig. 4.

and directions distributed between southwest and down to (clockwise) northeast and down (Fig. 8). Each sample was found to carry up to three NRM components with characteristic stability L_S: <200 °C or <5–10 mT; I_S: 100–580 °C or 5–50 mT; H_S: 580–680 °C.

Samples of all lithologies carried component L_S. The remanence component is directed steeply downward in approximate alignment with the present Earth's field and is regarded as recent VRM.

Component I_S was also carried by both red and green sediments. I_S is directed towards the south–southwest with shallow upward inclinations. Site level I_S component directions are given in Fig. 10(a) and Table 6.

Well-resolved H_S components were only derived from seven red siltstone samples. Of these only four, from sites S2 and 3 are consistent in direction; to the northeast at intermediate downward inclinations (Fig. 10b).

THE FOLD TEST

Tectonic models for the evolution of the Hornelen basin suggest that the present dip of strata might have been imposed diachronously across the basin from west (Hovden) to east (Storevatnet) in mid-late Devonian times. As the depocentre migrated, so tilting progressed eastward. The variation in bedding attitude within each sampling region is small (Tables 5 and 6), yet the difference between the average strike and dip of strata at Hovden (020°/33°E) and Storevatnet (350°/30°E) is sufficient to perform a fold test on equivalent remanence components from the two regions (that is I_H–I_S and H_H–H_S). The 'unfolding' of palaeomagnetic results is performed by rotating them about the local bedding strike.

Component *I*

The 'unfolding' loci of site level I_S (Storevatnet) directions and group of sites level I_H (Hovden) directions are given in Fig. 11(a). There is an immediate and continued divergence of the directions from the two regions during 'unfolding'. This divergence is not statistically significant (Storevatnet data are sparse) and therefore the fold test is inconclusive.

Although few in number, *in situ* I_S directions are consistently shallower than I_H (Fig. 11a). It seems that there cannot be a simple tectonic explanation for this. The anomaly is discussed further below.

Component *H*

A number of H_H directional trends were identified (Fig. 10b). The strata in the vicinity of sampling sites are homoclinal and therefore a tectonic explanation for the directional dispersion can be rejected. The most significant of the four main H_H directional trends, on a regional scale, is that defined by samples from sites H1, 2, 4, 5 and 7 (one, seven, two, one and three samples respectively). Perhaps significantly, this northeasterly shallow inclination trend (H_H) is similar to the only confirmed trend from Storevatnet (H_S). Clearly there are too few H_H and H_S results, and the level of distortion of the strata is too low to permit a reliable fold test of the *H* component stability.

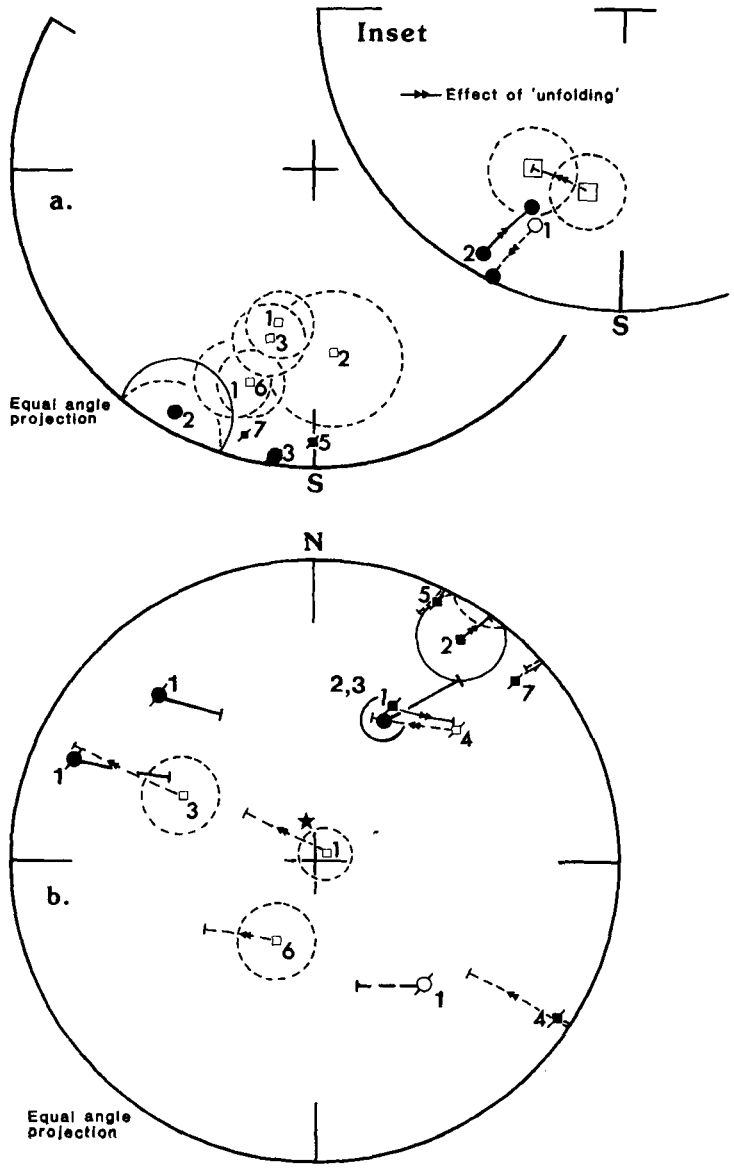


Figure 10. *I* component (a) and *H* component (b) directions from the Hornelen basin at site and (inset) site and group of sites level. Small (resp. large) squares denote Hovden results at site (resp. group of sites) level. Circles denote Storevatnet results at site level. Oblique strokes indicate mean directions which are not statistically significant. Arrows show the effect of 'unfolding'. Site numbers are shown. The star lies on the present Earth's field direction.

REGIONAL COMPARISON OF REMANENCE COMPONENT DIRECTIONS

Fisher mean $H_{E,N,H,S}$ and $I_{E,N,H,S}$ directions, at regional—group of sites—level where possible, are given in Figs 11(a) and (c) respectively. These trends are the best estimates of the regional NRM component directions.

Components $I_{E,N,H,S}$

The 95 per cent confidence circles about the results (except Hovden: I_H), overlap each other when regarded *in situ* (Fig. 11c) and become significantly more dispersed when strata are restored to the horizontal (Fig. 11d). The significant local I_E fold test indicates that I_E at Etrefjell is a post-/late tectonic overprint. The above comparison is consistent with this interpretation and implies that the post-tectonic overprint extends across both the Kvamshesten (Etrefjell and Norddalstjern districts) and Hornelen basins (Hovden and Storevatnet districts). Furthermore, this demonstrates that no substantial post-overprinting relative rotation of the two basinal sequences has occurred.

Component $H_{E,N,H,S}$

$H_{H,S}$ (Hornelen) directions are of normal polarity, and $H_{E,N}$ (Kvamshesten) of reversed polarity with respect to the generally accepted apparent polar wander (APW) path for Baltica. Reversed polarity counterparts to the Hornelen results ($H_{H,S}$) are given in Fig. 11 for easy comparison with

Table 5. Site and group of sites level statistics for the *H* and *I* components, Hovden.

H component

Site	Approx. Strat Positn/*m*	Bedding Strike	Dip	Remanence Dec.	Inc.	*N*	*k*	α_{95}
Top of formation	73							
5	72	016	29ESE	024.8	2.7	1	N.S.	—
1	72	025	33ESE	050.1	−86.0	3	137.5	10.6
				020.7	32.3	1	N.S.	—
6	67.5	016	29ESE	206.1	−57.6	6	23.8	14.0
3	67	018	37ESE	297.0	−38.2	6	28.2	12.8
2	66	024	37ESE	032.4	7.1	7	35.2	10.3
7a/b	20–32	020 26ESE/009 31E		047.3	6.5	3	11.5	38.3
4	23	023	33ESE	123.8	3.8	2	N.S.	—
Bottom of Fm.	0							

I component

Site	Remanence Dec	Inc.	*N*	*k*	α_{95}
5	181.5	4.8	1	N.S.	—
1	192.5	−34.6	5	59.9	10.0
6	196.7	−16.2	6	73.1	7.9
3	195.3	−29.1	6	40.1	10.7
2	175.3	−26.4	7	11.8	18.3
7a/b	177.7	−34.3	2	76.2	29.0
4	195.4	5.6	1	N.S.	—

Overall statistics (*I* component)

Using Fisher mean directions from sites K1, 2, 3, 6, 7a/b

Mean Direction:	Declination	Inclination	*N*	*R*	*k*	α_{95}	Long.	Lat.	VGP *dm*	*dp*	*Lp*
in situ	187.7	−28.4	5	4.9	47.9	11.2	354.6E	−43.1	12.3	6.7	15.1N ϕ
'Unfolded'	207.6	−30.8	5	4.9	38.7	12.5	328.9E	−40.8	13.9	13.9	16.6N

Notation as in Tables 1, 4; ϕ = VGP included in Fig. 12. *In situ* and 'unfolded' *H* and *I* component directions are given in Fig. 10.

Table 6. Site level *H* and *I* component directions, Storevatnet (*in situ*).

Site	Approx. Strat† Positn/*m*	Bedding Strike	Dip	Remanence *H* Dec.	Inc.	*B*	*k*	α_{95}	Remanence *I* Dec.	Inc.	*N*	*k*	α_{95}
				318.0	16.3	1	N.S.	—	201.3	−15.3	9	27.9	9.9
1	200	350	30	139.0	−32.5	1	N.S.	—	21.7	−39.8	1	N.S.	—
				293.9	8.9	1	N.S.	—					
2	60	351	28						208.5	3.6	5	33.2	13.5
				026.9	34.4	4	217.3	6.3					
3a/b	0–40	358	29						187.3	1.1	2	N.S.	—

Notation as in Table 1. † = with respect to S3. *H* and *I* component directions are given in Fig. 10.

results from Kvamshesten ($H_{E,N}$). 'Unfolded' *H* component directions seem to be better grouped than *in situ* ones (Fig. 11b and a resp.); Fisher's (1953) *k* for the two distributions are 18.6 and 11.4 respectively. Of course, with a sample of four ($N = 4$) a four-fold rise in *k* upon 'unfolding' is required for statistical significance. Therefore this comparison does not clarify the tentative $H_{E,N}$ fold test local to the Kvamshesten basin. The regional consistency in $H_{E,N,H,S}$ directions suggests that the remanences might all be of approximately the same age, yet data from other studies must be considered in an attempt to constrain that age further.

MAGNETIC MINERALOGY OF THE DETRITUS

Polished thin section work clearly indicates the presence of haematite in most specimens. That haematite takes the form of both specularite and pigment. Haematite is further indicated by Curie balance experiments (T_c = *c.* 660–690 °C) and isothermal remanent magnetization (IRM) acquisition curves (saturation is not reached by 1.2 T, the maximum field available). The coercivity of remanence in the redbeds is at least 80 mT. The high and narrow unblocking temperature spectrum of *H* (Fig. 4) and its high coercivity

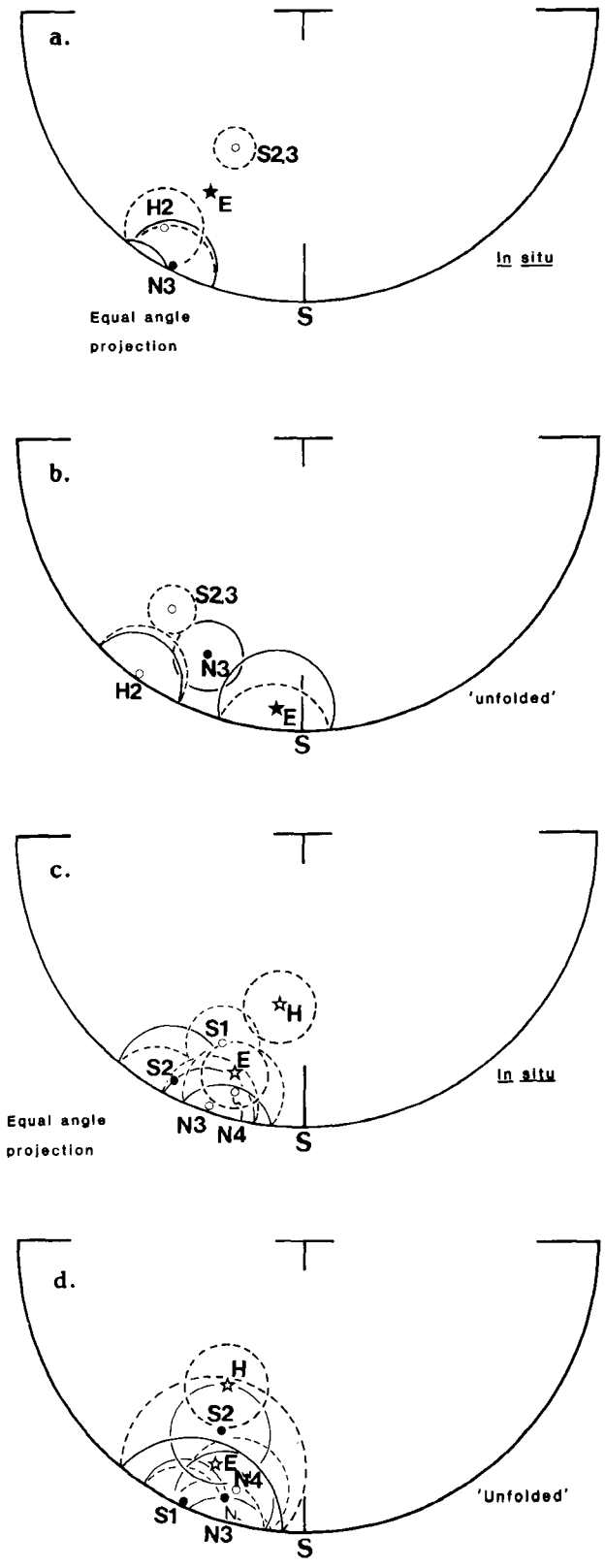


Figure 11. Remanence component directions for the Hornelen and Kvamshesten basins. (a) *H in situ*, (b) *H* 'unfolded', (c) *I in situ* and (d) *I* 'unfolded'. Where possible group of sites level result (stars) are shown. Site level results are denoted by circles. H: Hovden, S: Storevatnet, E: Etrefjell, N: norddalstjern. Site numbers are shown where appropriate.

unambiguously point to a haematite host. Whether the remanence resides in the specularite or post-depositional (diagenetic) pigment, or both, is uncertain.

Curie balance experiments on redbed samples indicate a further mineral phase with T_c in the region 550°–600 °C. Moreover, IRM acquisition curves show saturation of a mineral phase in fields of *c.* 0.1 T. These characteristics are consistent with a magnetite host. The observed T_b spectra in the magnetite stability range are broad and it is presumed that the (magnetite) grains are small and single domain. A number of euhedral magnetite grains have been identified in both the red and green sediments indicating a secondary origin for at least some of the magnetite. Remanence I is stable above the Curie point of magnetite and is presumed to be carried by both magnetite and haematite.

Samples which were subjected to heating in the Curie balance exhibited a more pronounced magnetite Curie point during cooling than heating. This might be due to Pyrrotite inverting to magnetite at *c.* 350–450 °C, or reduction of haematite to magnetite at around 500–600 °C.

PREVIOUS PALAEOMAGNETIC RESULTS FOR THE KVAMSHESTEN AND HORNELEN BASINS

Palaeomagnetic data from the sediments of the Kvamshesten basin have been reported by Lie, Storetredt & Gjellestad (1969). Their analysis incorporated the removal of a low-stability NRM component, leaving the vector structure of the remaining remanence unclear. The present study implies that the sediments in the interior part of the basin carry two NRM components (*I* and *H*) stable above 300 °C. Incomplete resolution of these two characteristic remanence components might explain the large directional scatter in their data, and account for their inconclusive fold test. This notion is corroborated by the fact that their data (referred to bedding) are distributed between the characteristic 'unfolded' $H_{E,N}$ and $I_{E,N}$ trends of this study.

Further palaeomagnetic data for the Kvamshesten massif have been provided by Torsvik *et al.* (1986). Much of their sampling was confined to sediments in the vicinity of the basal detachment and areas exhibiting cleavages, contrary to the sampling strategy adopted in this study. Consequently, and quite expectedly, there are some substantial differences in the NRM characteristics of samples obtained in their and this study. Sites 47 to 53 of Torsvik *et al.* (1986) were made in the vicinity of sites E1 to 11 of this study. Their analysis revealed a vector component ('*A*'; Pole KA in Fig. 12) with unblocking temperatures exceeding 300 °C. Examination of their *in situ* remanence component directions in this stability range (their fig. 7) and the *in situ* component I_E trend and the predominant H_E trend of this study [*I*: 150° to 650 °C, fig. 5(a); *H*: 610°–680 °C, fig. 6] suggests that, in the interior part of the basin, component '*A*' is likely to be an amalgamation of H_E (southeast and down) and the better defined I_E (south–southeast and up). Examination of their fiig. 7(b) suggests that an equivalent remanence to $H_{E1,2,5,8,9}$ was identified at site 48. I_E is well defined and evidently post-/late folding, and $H_{E,N}$, although ambiguous at a number of sites, could be early/pre-folding. If I_E and $H_{E,N}$ were considered together, or not resolved, 'unfolding' might result in comparatively little change in the overall

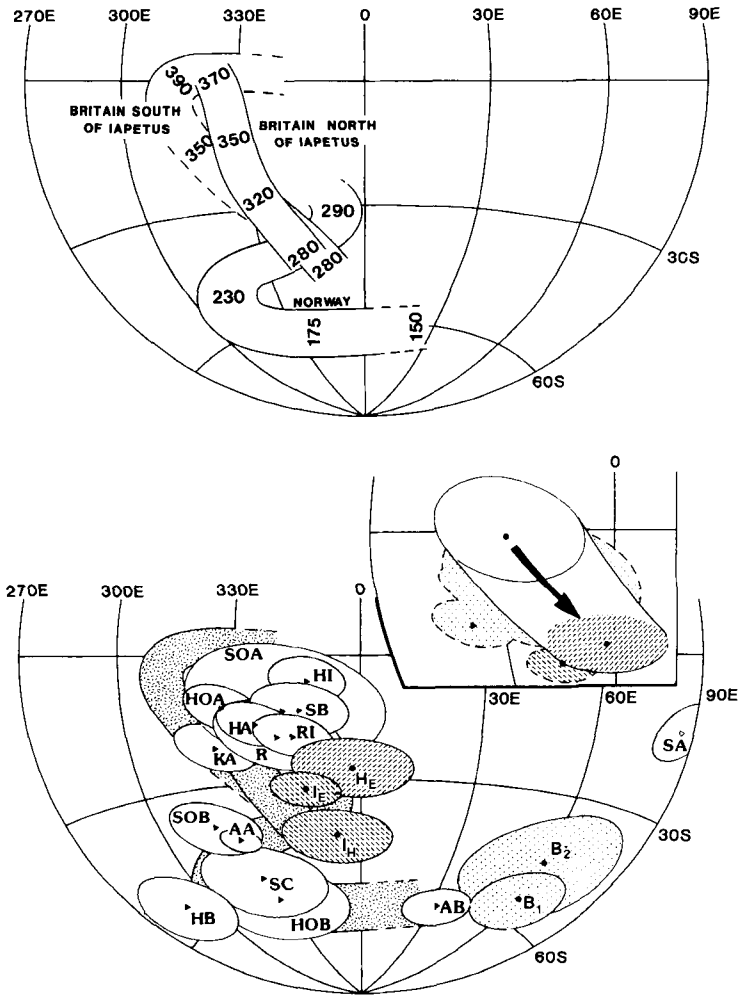


Figure 12. Top: reference APW paths for Britain (Torsvik *et al.* 1989b) and Norway. Ages are in Myr BP. Bottom: as Top overlain by palaeomagnetic poles from mid-Palaeozoic rocks in Norway. Circles (and shaded error ovals): this study. KA, KB: Kvamshesten; HOA, HOB: Hornelen; HA, HB: Håsteinen; SOA, SOB: Solund; R: Røragen; RI: Ringerike; SA, SB, SC: Smøla; AA, AB: Atløy. Inset: effect of 'unfolding' on H_E. Error limits are of 95 per cent confidence. For source references and further explanation see text.

mean direction and Fisher precision, as observed by Torsvik *et al.* (1986). Torsvik (op cit.) recognize possible contamination of their results with a young 'Permian (?)' overprint, weakening their argument for a syn-folding (30 per cent) magnetization event. That overprint is isolated in this study and is labelled *I*. Notwithstanding this, the '*A*' component of Torsvik *et al.* (1986) from (local to) the Devonian mylonites at the margin of the basin is well defined, stable up to the Curie point of haematite, and clearly secondary in origin. It is supposed that inferences drawn from those data are of the highest integrity.

Palaeomagnetic data are also available for the sediments of the Hornelen basin (Torsvik *et al.* 1989c). Their '*B*' component (their fig. 15; Pole HOB in Fig. 12) is probably correlative with $L_{H,S}$ and possibly the $H_{H1,3,6}$ reversed polarity site mean directions of this study (*H*: fig. 10b). $H_{H,S}$ of this study might correspond with their '*A*' component (HOA in fig. 12), both of which are dual polarity, however no obvious counterpart to $I_{H,S}$, the most frequently isolated remanence in this study, was regularly forthcoming in the study of Torsvik *et al.* (1989c). The most plausible explanation for this is that the current study is largely based on analysis of samples from subordinate redbeds situated where possible, away from the tectonic margins of the basin (western Hovden and Storevatnet), whereas all but two sites of Torsvik (op cit.) were in grey sand/siltstones, some along the margins of the basin, or in basement rocks.

THE PALAEOZOIC APW PATH FOR BALTICA

The Middle and Upper Palaeozoic section of the APW path for Baltica is poorly constrained by well-dated palaeomagnetic data with satisfactory tectonic control from Scandinavia. Furthermore, there are few strong constraints on the age of much of the data from the Russian Platform (Smethurst, Khramov & Briden 1989).

In a recent review by Torsvik *et al.* (1989b) Palaeozoic palaeo-positions for Baltica were determined using data from Scandinavia alone. That study, like a number of others, showed that there are too few reliably dated results for mid to late Palaeozoic time which can accurately define

APW for Baltica. Therefore the new data from this study are compared with selected individual palaeopole positions for Baltica (Fig. 12), and the better constrained Palaeozoic APW paths for Britain of Torsvik *et al.* (1989b); and with the Mesozoic path for Baltica of Torsvik *et al.* (1989d; Fig. 12 top). The palaeomagnetic data from which the paths were statistically derived are listed in the two respective papers. The list of British data is an update on those data compiled by Livermore *et al.* (1985) and Briden *et al.* (Briden, Turnell & Watts 1984; Briden *et al.* 1989).

A number of palaeomagnetic pole positions have been obtained of variable in reliability for the Old Red Sandstone of Norway. Palaeopoles reported, in some cases tentatively, as Devonian in age include those from the Kvamshesten sediments (Lie *et al.* 1969) and sediments/mylonites (Torsvik *et al.* 1986; KA in Fig. 12); Håsteinen sediments and basement (Torsvik *et al.* 1987; Fig. 1; HA); Røragen sediments (Storetvedt & Gjellestad 1966; Fig. 1; R); Hornelen sediments and fault rocks (Torsvik *et al.* 1989c; HOA); Smøla sediments and basement (Torsvik *et al.* 1989d; Fig. 1—Trondheimsfjord; SA, SB); Hitra sediments (Bøe, Atakan & Sturt 1989; fig. 1—Trondheimsfjord; HI); Solund sediments (Torsvik *et al.* in preparation; Fig. 1; SOA). A Silurian result for the Ringerike sediments and related rocks is also shown in Fig. 12 (RI; fig. 1; Douglass 1988). Early Mesozoic overprints were reported in the Solund (SOB) and Atløy (AA) studies, and later Mesozoic overprints in the Håsteinen (HB), Hornelen (HOB), Smøla (SC) and Atløy (AB) studies.

The cluster of results labelled HOA, SOA, HI, SB, and HA from the Old Red Sandstone and basement are thought by the respective authors to post-date late Caledonian (Solundian) folding but still be Devonian in age from comparison with British poles. KA is interpreted as syn-late tectonic and is hence assigned a Devonian age by Torsvik *et al.* (1986). There is little independent age control on the R pole. The reporters of SOB, AA, SC, HB, HOB and AB presume them to be of Mesozoic age from comparison with reference data, and the inference that the faults with which some of them seem to be associated were probably active in the Mesozoic.

Comparison of Scandinavian and British data suggests that the two continental units had, within error bounds, their present latitudinal configuration in Devonian time, but that the palaeomeridian for Scandinavia was nearer to the present meridian than that for Britain. This can be accounted for by either (1) late/post-Devonian relative rotation between Scandinavia and Britain or (2) inadequate resolution of the APW path for Britain, or (3) inadequate dating/structural control on a number of the Scandinavian data. The first possibility is not favoured because there is little evidence for very substantial late/post-Devonian transcurrent motion between Britain and Norway. The second, non-tectonic theory is perhaps the most plausible. Few of the data from Norwegian sediments are interpreted as primary, hence the discrepancy might arise from differential remagnetization. Nevertheless, the Lower Devonian and Permian segments of the British APW path are comparatively well defined, but the existing data for Britain do not preclude some west to east APW from Silurian through to Devonian time, taking the British path near to/through the Devonian Old Red Sandstone data from Norway. This possibility, then, requires north to south APW through Upper Palaeozoic time.

Component B_N

Pole positions corresponding to remanence B_N at Kvamshesten basin sites N2 and 3 are given in Fig. 12 (B_1 and B_2). The high latitudes for B_N *in situ* pole positions—and local, qualitative, negative fold test—suggests that B_N is Mesozoic or younger in age. Although the Mesozoic to present APW for Baltica is not well time calibrated the site level B_N poles border on the extrapolated Jurassic/early Cretaceous segment of the reference APW path. Therefore B_N is taken to be Mesozoic—probably Jurassic/early Cretaceous—in age.

Component *I*

Palaeopoles corresponding to the post-/late folding Kvamshesten I_E and Hornelen I_H group of sites level results fall on the late Carboniferous/early Permian part of all reference paths (I_E, I_H: Fig. 12). Therefore it is taken that the region wide overprint is latest Palaeozoic/early Mesozoic in age.

Component *H*

A local positive fold test in the Kvamshesten basin and comparison of $H_{E,N}$ component directions from Kvamshesten and $H_{H,S}$ from Hornelen suggests that $H_{E,N,H,S}$ has a pre- or early folding age and have significance 'unfolded'. Nevertheless, the pole position corresponding to H_E ('unfolded') is removed from the loose cluster of 'Devonian' reference palaeopoles for Baltica (H_E: Fig. 12), and further removed from the reference paths for Britain.

The Devonian palaeolatitude for western Norway implied by the new result 'unfolded' (resp. *in situ*) is 1.9°S ± 13.9° (resp. 24.0 °S ± 18.5°) and fixes Devonian Norway at 5°–10° (resp. 12°–17°) north (resp. south) than its present position relative to Britain. It should be emphasized that these implied latitudinal separations are not statistically significant. It is notable that there is no significant discrepancy in palaeomeridian between the *in situ* H_E result and British data, and 'unfolding' the H_E result causes it to diverge from the British Devonian reference data (Fig. 12 inset).

Clearly the aforementioned complexities in relative palaeopole positions have implications for several possible geological models. In palaeomagnetic terms either (1) *H* pre-dates folding and the corresponding pole position lies near to the Devonian palaeopoles for Baltica, but is at variance with the paths for Britain, (2) *H* is syn/post-folding, corresponding to a pole in close agreement with the Baltica data and near to some British results (Fig. 12: inset) or (3) that the strata were not horizontal when they were magnetized (*H*).

The first possibility perpetuates the longstanding discrepancy between British and Norwegian Devonian data. This contradicts geological evidence, which denies substantial latitudinal separation of Britain and Fennoscandia in

Devonian time. It also implies a substantial rotational component to the movement between the two in/since Devonian time.

The second possibility only becomes inconsistent with the local positive fold test when the remanence is taken to post-date much of the distortion of the strata. Attention, here, should be drawn to the controlling role of a single site level result in the (H_E) fold test.

The third possibility has important implications with regard to the early development of the Devonian basins. For example, if an easterly tilt was imposed on some/all of the Kvamshesten strata prior to remanence acquisition (H) and folding the 'corrected' palaeofield is left with a more characteristic (of the Devonian of Britain) south-southwest to southwest azimuth. This is permitted by the fold test as applied in the Kvamshesten basin.

CONCLUSIONS

The multicomponent structure of the NRM carried by the sediments in the interior part of the Kvamshesten basin has been effectively demonstrated for the first time. Similar multicomponent properties of NRM carried by rocks in the Hornelen basin were also resolved. The Devonian sediments at almost all sampling sites in the Hornelen and Kvamshesten basins carry a well-defined post-tectonic reversed polarity remanence component. The highest precision regional mean direction for this remanence is from the Kvamshesten basin, and has declination 196°, inclination 8.8°, α_{95} 8.0° which corresponds to a south pole position at longitude 346.6 °E, latitude −31.7 °N with *dp/dm* 4.1/8.1° (palaeolatitude 4.4 °N). A second, probably post-tectonic, remanence of similar stability was obtained from the western end of the Hornelen basin, with dec. 187.7°, inc. −28.4°, α_{95} 11.2°, long. 354.6 °E, lat. −43.1 °N, *dp/dm* 6.7/12.3° (plat. 15.1 °N).

The best estimate of the ages of these overprints comes from palaeopolar comparison with reference data sets for Britain and Norway, which dates them at late Palaeozoic and late Palaeozoic/early Mesozoic respectively. These are among very few well-resolved regional overprints carried by the Devonian Old Red Sandstone of Norway which could be of the 'Kiaman' genera, more frequently identified elsewhere on the 'Old Red Continent'. Perhaps uncharacteristically, the ORS of Norway is reported to carry Devonian/Lower Carboniferous and Mesozoic (Jurassic?) overprints.

Consistent with this observation, a second, local, magnetic overprint (CRM) of probable Mesozoic (Jurassic) age was identified in the vicinity of the high-angle E–W Grundvatn Fault. The remanence effectively records and dates a later phase of thermochemical activity and mineralogical alteration associated with, and local to, the fault. This observation corroborates the arguments of Torsvik *et al.* (1989c) for Mesozoic faulting and fault reactivation in western Norway, probably linked with lithospheric extension in the North Sea (White 1989). Indeed, this fault activity might be responsible for extensive magnetic resetting in rocks both in western Norway and Scotland.

The locus of Silurian to Carboniferous APW for Baltica remains controversial. Conventional geological wisdom suggests that at least post-Silurian APW for Norway should coincide with that for Britain. In the literature differences between the paths have been attributed to uncertainty in their time calibration. This explanation is to some extent undermined by the fact that most of the Norwegian Devonian data lie to one side of the British APW path, not on a younger part of it. The explanation for this difference might be tectonic. Otherwise, and more probably, the two data sets might lie on a single but more complicated/tightly curved APW path than that offered here.

Most redbed samples were found to carry remanence with high stability, probably carried by diagenetic haematite pigment. Although comparatively few reliable site level directions for this NRM component were forthcoming in the Kvamshesten basin, a statistically significant fold test ($p = 0.01$) implies an early/pre-folding age. However, the test was applied following (necessary) data selection and depends on a single site level result, and therefore has a low geological significance.

The associated *in situ* (resp. 'unfolded') palaeopole has long. 335.7 °E (resp. 359.4 °E), lat. −0.2 °N (resp. −26.5 °N) with *dp/dm* 11.4/18.5° (resp. 7.0/13.9°). The two palaeopoles indicate palaeolatitudes for western Norway of 24 °S ± 18.5° and 1.9 °S ± 13.9° respectively, which, bearing error bounds in mind, are consistent with Norway and Britain attaining their present latitudinal configuration by mid-Devonian time. If the fold test has significance then the 'unfolded' result provides an anomalous Devonian palaeomeridian for Norway trending N–S, rather than an expected SE to S–SE trend. This phenomenon might be explained in terms of (1) local late/post-Devonian rotation of the Kvamshesten basin, or (2) a syn-folding age for the remanence or (3) syn-depositional easterly tilting of Kvamshesten sediments before they were magnetized and folded. In the latter case, a positive fold test suggests that the sediments were folded after they acquired an easterly tilt, and hence that tilting and folding were not entirely contemporaneous.

Although there is a paucity of reliable independent evidence for the age of this remanence in relation to folding, the most plausible inference, on the ground of simplicity and conformity with reference data, is that the remanence post-dates much of the distortion of the strata (and is later Devonian/Lower Carboniferous). The *in situ* pole position, then, is believed to have geological significance, and is a major revision on the previous results for the Kvamshesten basin. Implicit to this interpretation, and the interpretations of aforementioned recent authors, is the possibility that no early-folding (primary?) remanence might survive in the Devonian of western Norway. Further detailed palaeomagnetic analysis is required to test this theory.

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REFERENCES

Briden, J. C., Turnell, H. B. & Watts, D. R., 1984. British palaeomagnetism, Iapetus ocean, and the Great Glen Fault, *J. Geol.*, **12**, 428–431.

Briden, J. C., Kent, D. V., Lapointe, P. L., Livermore, R. A., Roy, J. L., Seguin, M. K., Smith, A. G., Van der Voo, R. & Watts, D. R., 1989. Palaeomagnetic constraints on the evolution of the Caledonian–Appalachian orogen, in *The Caledonian–Appalachian Orogen*, pp. 35–48, eds. Harris, A. L. & Fettes, D. J., Geol. Soc. Spec. Publ. No. 38, London.

Bryhni, I., 1964. Migrating basins on the Old Red Continent, *Nature*, **202**, 384–385.

Bryhni, I. & Skjerlie, F. J., 1975. Syndepositional tectonism in the Kvamshesten district (Old Red Sandstone), western Norway, *Geol. Mag.*, **112**, 593–600.

Bøe, R., Atakan, K. & Sturt, B. A., 1989. The style of deformation of the Devonian rocks of Hitra and Smola, *Nor. geol. Tidsskr.*

Chauvet, A. & Brunel, M., 1988. La grande faille normale ductile du Sunnfjord. Une extension NW–SE dans la chaîne Calédonienne, *C. R. Acad. Sci. Paris*, **307**, 415–422.

Chauvet, A. & Séranne, M., 1988. Microtectonic evidences of Devonian extensional westward shearing in southwest Norway, in *The Caledonian and Related Geology of Scandinavia*, ed. Gayer, R., Graham & Trotman, London.

Douglass, D. N., 1988. Palaeomagnetics of Ringerike Old Red Sandstone and related rocks, southern Norway: implications for pre-Carboniferous separation of Baltica and British terranes, *Tectonophysics*, **148**, 11–27.

Everitt, C. W. F. & Belshe, J. C., 1960. Palaeomagnetism of the British Carboniferous system, *Phil. Mag.*, **5**, 675–685.

Fisher, R., 1953. Dispersion on a sphere, *Proc. R. Soc. Lond.*, A, **217**, 295–304.

Hoeg, O. A., 1936. Norges fossile flora, *Naturen*, **7–21**, 47–64.

Hossack, J. R., 1984. The geometry of listric normal growth faults in the Devonian basins of Sunnfjord, western Norway, *J. geol. Soc. Lond.*, **141**, 629–637.

Høisethær, T., 1971. Thrusted Devonian sediments in the Kvamshesten area, western Norway, *Geol. Mag.*, **108**, 287.

Jarvik, E., 1949. On the Middle Devonian crossopterygians from the Hornelen field in western Norway, Universitet i Bergen Aarbok, *Naturvidensk*, **8**, 1–48.

Kent, J. T., Briden, J. C. & Mardia, K. V., 1983. Linear and planar structure in ordered multivariate data, as applied to progressive demagnetisation of palaeomagnetic remanence, *Geophys. J. R. astr. Soc.*, **75**, 593–621.

Khramov, A. N., Petrova, G. N. & Pechersky, D. M., 1981. Palaeomagnetism of the Soviet Union, in *Palaeoreconstruction of the Continents*, Geodynamic Series, Vol. 2, 177–194, American Geophysical Union, Washington, DC.

Kiaer, J., 1918. Fifkerester fra den devonske sandsten paa Norges Vestkyst, (Engl. summ.) Bergens Mus. Aarbok, *Naturvidensk*, **7**, 1–17.

Kisch, H. J., 1987. Correlation between indicators of very low grade metamorphism, in *Low Temperature Metamorphism*, chapter 7, pp. 227–300, ed. Frey, M., Blackie and Son, Glasgow.

Kolderup, C. F., 1915. Vestlandets devonfelter og deres plantefossiler, *Naturen*, 217–232.

Larsen, V. & Steel, R. J., 1978. The sedimentary history of a debris flow-dominated alluvial fan—a study of textural inversion, *Sedimentology*, **25**, 37–59.

Larsen, V., Spinnager, A., Steel, R. J., Aasheim, S., Gloppen, T. G. & Maehle, S., 1981. A Field Guide to Hornelen Basin—A Deep Late Orogenic Basin (Devonian) in western Norway, *Internal Report*, University of Bergen.

Lie, L. G., Storetvedt, K. M. & Gjellestad, G., 1969. The palaeomagnetism of the Kvamshesten Old Red sequence, southwest Norway, *Nor. geol. Tidsskr.*, **49**, 241–249.

Livermore, R. A., Smith, A. G. & Briden, J. C., 1985. Palaeomagnetic constraints on the distribution of continents in the late Silurian and early Devonian, *Phil. Trans. R. Soc. Lond.*, B, **309**, 29–56.

Løvlie, R., Torsvik, T. H., Jelenska, M. & Lavandowski, M., 1984. Evidence for detrital remanent magnetisation carried by haematite in Devonian red beds from Spitzbergen: palaeomagnetic implications, *Geophys. J. R. astr. Soc.*, **79**, 573–588.

Norton, M. G., 1986. Late Caledonian extension in western Norway: a response to extreme crustal thickening, *Tectonics*, **5**, 195–204.

Padan, A., Kisch, H. J. & Shagam, R., 1982. Use of the lattice parameter b_0 of dioctahedral illite/muscovite for the characterisation of P/T gradients of incipient metamorphism, *Contrib. Miner. Petrol.*, **79**, 85–95.

Pesonen, L., Torsvik, T. H., Bylund, G. & Elming, S. A., 1989. Crustal evolution of Fennoscandia—Palaeomagnetic constraints, *Tectonophysics*, **162**, 27–49.

Pollard, J. E., Steel, R. J. & Undersrud, E., 1982. Facies sequences and trace fossils in lacustrine/fan delta deposits, Hornelen Basin (M. Devonian), western Norway, *Sedim. Geol.*, **32**, 63–87.

Roberts, D., 1983. Devonian tectonic deformation in the Norwegian Caledonides and its regional perspectives, *Nor. geol. Unders.*, **380**, 85–96.

Robinson, M. A., 1985. Palaeomagnetism of volcanics and sediments of the Eday Group, southern Orkney, *Scott. J. Geol.*, **21**, 285–300.

Scotese, C. R., van der Voo, R. & Barrett, S. F., 1985. Silurian and Devonian base maps, *Phil. Trans. R. Soc. Lond.*, B, **309**, 57–77.

Séguret, M., Séranne, M., Chauvet, A. & Brunel, M., 1989. Collapse basin: a new type of extensional sedimentary basin from the Devonian of Norway, *Geology*, **17**, 127–130.

Smethurst, M. A. & Khramov, A. N., 1989. New palaeomagnetic data from the Devonian of the Russian Platform: its bearing on the problem of differential remagnetisation in Europe, in *Book of Abstracts*, IAGA-6, Exeter 24th July–4th August, 1989 (abstract).

Smethurst, M. A., Khramov, A. N. & Briden, J. C., 1989. New palaeomagnetic data for Siluro-Devonian sediments in Podolia, USSR, *Terra*, **1**, 82 (abstract).

Steel, R. J., 1976. Devonian basins of western Norway—sedimentary response to tectonism and varying tectonic context, *Tectonophysics*, **36**, 207–224.

Steel, R. J. & Aasheim, S., 1978. Alluvial sand deposition in a rapidly subsiding basin (Devonian, Norway), in *Fluvial Sedimentology*, vol. 5, pp. 385–413, ed. Miall, A. D., Can. Petrol. geol. Mem., Calgary.

Steel, R. J. & Gloppen, T. G., 1980. Late Caledonian Devonian basin formation, western Norway: signs of strike-slip tectonics during infilling, *Spec. Publ. int. Assoc. Sediment.*, **4**, 79–103.

Steel, R. J., Siedlecka, A. & Roberts, D., 1985. The Old Red Sandstone basins of Norway and their deformation: a review, in *The Caledonide Orogen—Scandinavia and Related Areas*, pp. 293–315, eds Gee, D. G. & Sturt, B. A., John Wiley & Sons, Chichester.

Steel, R. J., Maehle, S., Røe, S., Spinnager, A. & Nilsen, H. R., 1977. Coarsening-upward cycles in the alluvium of the Hornelen Basin (Devonian, Norway)—sedimentary response of the tectonic events, *Bull. geol. Soc. Am.*, **88**, 1124–1134.

Storetvedt, K. M. & Gjellestad, G., 1966. Palaeomagnetic investigation of an Old Red Sandstone formation of southern Norway, *Nature*, **212**, 59–61.

Stubbs, P. H. S., 1958. Continental drift and polar wandering, a palaeomagnetic study of the British and European Triassic and British Old Red Sandstone, *PhD thesis*, University of London.

Sturt, B. A., 1983. Late Caledonian and possible Variscan stages in the orogenic evolution of the Scandinavian Caledonides, in *The Caledonian Orogen-IGCP Project 27,* Symposium de Rabbat (abstract), Morocco.

Thorning, L., 1974. Palaeomagnetic results from Lower Devonian rocks of the Cheviot Hills, northern England, *Geophys. J. R. astr. Soc.*, **36,** 489–496.

Torsvik, T. H., Holme, J., Sturt, B. A. & Vetti, V., 1987. The tectono-magnetic signature of the Old Red Sandstone and pre-Devonian strata in the Håsteinen area, Western Norway, and implications for the later stages of the Caledonian orogeny, *Tectonics,* **6,** 3305–3322.

Torsvik, T. H., Smethurst, M. A., Briden, J. C. & Sturt, B. A., 1989b. A review of palaeomagnetic data from Europe and their palaeographical implications, in *Palaeozoic Biogeography and Palaeobiogeography,* eds M. Kerrow, W. S. & Scotese C. R., Mem. geol. Soc. Lond., in press.

Torsvik, T. H., Sturt, B. A., Ramsay, D. M., Bering, D. & Fluge, P. R., 1989c. Palaeomagnetism, magnetic fabrics and the structural style of the Hornelen Old Red Sandstone, Western Norway, *J. geol. Soc. Lond.*, **145,** 413–430.

Torsvik, T. H., Sturt, B. A., Ramsay, D. M., Kisch, H. J. & Bering, D., 1986. The tectonic implications of Solundian (Upper Devonian) magnetisation of the Devonian rocks of Kvamshesten, western Norway, *Earth planet. Sci. Lett.*, **80,** 337–347.

Torsvik, T. H., Sturt, B. A., Ramsay, D.M., Gronlie, A., Roberts, D., Smethurst, M. A., Atakan, K., Boe, R. & Waldahaug, H. J., 1989d. Palaeomagnetic constraints on the early history of the Møre–Trøndelag fault zone, central Norway, in *Palaeomagnetic Rotations and Continental Deformation,* pp. 431–457, eds Kissel, C. & Laj, C., Kluwer Academic Publishers, Dortrecht.

White, N., 1989. Nature of lithospheric extension in the North Sea, *Geology,* **17,** 111–114.